

MAA-200 Split System - Installation Manual

Model MAA-200 | Rev A | 2019

IM-MAA-200

Prototype sample document for the clickable demo. Not an official publication.

Rev	Date	Description
A	2019-04	Initial release.
B	2021-02	Updated clearances and line-set tables.
C	2023-06	Electrical and control-wiring updates.

> For professional use only. Service work must be performed by qualified personnel.

Contents

Section	Page
Scope	3
Safety	4
Tools Required	5
Clearances	6
Mounting the Outdoor Unit	7
Refrigerant Line Sets	8
Brazing and Purge	9
Pressure Testing	10
Evacuation	11
Electrical Service	12
Low-Voltage Wiring	13
Thermostat Wiring	14
Control Wiring Diagram	15
Condensate Drainage	16
Startup and Checkout	16

> Page numbers refer to the printed page footer.

Safety

Section 3: Safety

All work must follow the applicable local and national codes. Only factory-approved replacement parts may be used, matched to the full part number including the revision suffix.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
Control signal	24 VAC	Call for operation
Ambient	see chart	Record at test

> Use the wiring diagram in Appendix D to confirm pinouts before probing.

Tools Required

Section 4: Tools Required

Component and connector locations are shown in the reference drawings. Confirm each pin against the drawing before probing or reconnecting any harness.

Procedure notes

- Use only factory-approved parts.
- Match the full part number and revision.
- Transfer configuration settings where fitted.
- Confirm routing against the reference drawings.
- Run a full startup sequence after assembly.

Terminal	Wire	Function
R	RED	24VAC supply
C	WHT	Common return
W	BLU	Call for heat
G	GRN	Call for fan

> Always confirm 0VAC and 0VDC before making resistance measurements.

Clearances

Section 5: Clearances

Torque all electrical terminations to the value shown in the specifications table. Loose terminations overheat and can damage the terminals over time.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize before recording.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not an assumption.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness

> Record the ambient and suction-line temperatures together for an accurate comparison.

Mounting the Outdoor Unit

Section 6: Mounting the Outdoor Unit

When a reading is out of range, do not replace the component until the associated wiring and connectors have been verified good. Most returned parts test good on the bench.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
Control signal	24 VAC	Call for operation
Ambient	see chart	Record at test

> If a fault returns after clearing, inspect the harness before replacing any board.

Refrigerant Line Sets

Section 7: Refrigerant Line Sets

Document the final readings and any parts used on the work order. Accurate records shorten the next visit and support the warranty claim.

Procedure notes

- Use only factory-approved parts.
- Match the full part number and revision.
- Transfer configuration settings where fitted.
- Confirm routing against the reference drawings.
- Run a full startup sequence after assembly.

Terminal	Wire	Function
R	RED	24VAC supply
C	WHT	Common return
W	BLU	Call for heat
G	GRN	Call for fan

> Use the wiring diagram in Appendix D to confirm pinouts before probing.

Brazing and Purge

Section 8: Brazing and Purge

This section covers the recommended field procedure. Follow the steps in order and verify each measurement before moving on. Record every reading on the commissioning worksheet.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize before recording.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not an assumption.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness

> Always confirm 0VAC and 0VDC before making resistance measurements.

Pressure Testing

Section 9: Pressure Testing

Before opening any panel, disconnect power at the service switch and wait for the DC bus to discharge. Confirm 0VAC and 0VDC at the board before proceeding.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
Control signal	24 VAC	Call for operation
Ambient	see chart	Record at test

> Record the ambient and suction-line temperatures together for an accurate comparison.

Evacuation

Section 10: Evacuation

Use a calibrated multimeter with a true-RMS function for all AC measurements. Resistance checks must be made with the unit de-energized and the harness disconnected.

Procedure notes

- Use only factory-approved parts.
- Match the full part number and revision.
- Transfer configuration settings where fitted.
- Confirm routing against the reference drawings.
- Run a full startup sequence after assembly.

Terminal	Wire	Function
R	RED	24VAC supply
C	WHT	Common return
W	BLU	Call for heat
G	GRN	Call for fan

> If a fault returns after clearing, inspect the harness before replacing any board.

Electrical Service

Section 11: Electrical Service

All work must follow the applicable local and national codes. Only factory-approved replacement parts may be used, matched to the full part number including the revision suffix.

Procedure notes

- Measure with the unit in the same mode as the fault.
- Allow readings to stabilize before recording.
- Note the ambient temperature at the time of test.
- Repeat any suspect measurement twice.
- Compare to the published range, not an assumption.

Code	Meaning	Action
E1	High pressure	Check charge and airflow
E2	Low pressure	Check charge and leak
E5	AC overcurrent	See Section 5.2
E6	Communication	Check harness

> Use the wiring diagram in Appendix D to confirm pinouts before probing.

Low-Voltage Wiring

Section 12: Low-Voltage Wiring

Component and connector locations are shown in the reference drawings. Confirm each pin against the drawing before probing or reconnecting any harness.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
Control signal	24 VAC	Call for operation
Ambient	see chart	Record at test

> Always confirm 0VAC and 0VDC before making resistance measurements.

Thermostat Wiring

Section 13: Thermostat Wiring

Torque all electrical terminations to the value shown in the specifications table. Loose terminations overheat and can damage the terminals over time.

Procedure notes

- Use only factory-approved parts.
- Match the full part number and revision.
- Transfer configuration settings where fitted.
- Confirm routing against the reference drawings.
- Run a full startup sequence after assembly.

Terminal	Wire	Function
R	RED	24VAC supply
C	WHT	Common return
W	BLU	Call for heat
G	GRN	Call for fan

> Record the ambient and suction-line temperatures together for an accurate comparison.

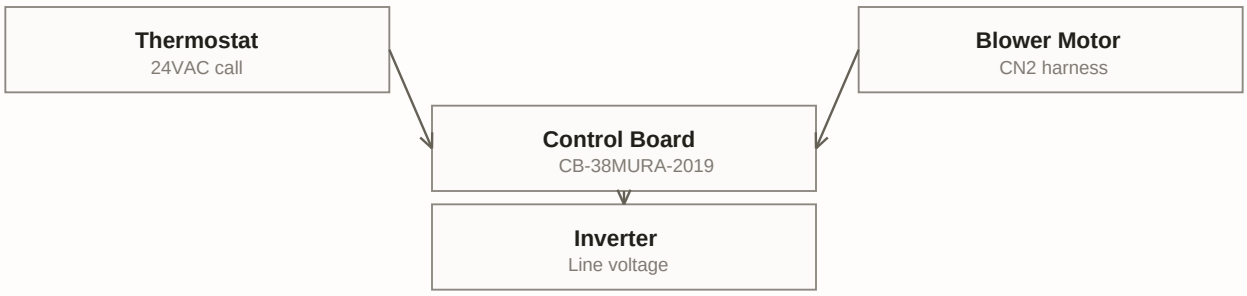
Control Wiring Diagram

Section 7.2: Wiring

Two-wire control connection (rev A).

Route the control wiring through the grommet on the left-hand panel. Connect the thermostat to terminals R and C. Diagram 7.2 shows a two-wire control connection.

Terminal	Wire	Function
R	RED	24VAC supply
C	WHT	Common return



Two-wire control connection - thermostat to control board

> This revision of the manual shows a two-wire arrangement. Confirm the engineering spec revision before wiring.

Condensate Drainage

Section 15: Condensate Drainage

Document the final readings and any parts used on the work order. Accurate records shorten the next visit and support the warranty claim.

Procedure notes

- Disconnect and lock out power.
- Confirm 0VAC and 0VDC.
- Inspect connectors for corrosion.
- Record baseline readings before any change.
- Reassemble and re-check operation.

Item	Expected	Notes
Line voltage	208-230 VAC	+/-10%
Low voltage	24 VAC	+/-10%
Control signal	24 VAC	Call for operation
Ambient	see chart	Record at test

> Use the wiring diagram in Appendix D to confirm pinouts before probing.

Startup and Checkout

Section 3.1: Startup

Power the unit and confirm the control board completes its startup sequence. Record any fault code before it clears.

Startup steps

- Confirm all electrical and refrigerant connections are complete.
- Energize the unit at the service switch.
- Observe the control board startup sequence.
- Record any fault code before it clears.

Code	Meaning	Reference
E1	High pressure	Service Manual 5.3
E2	Low pressure	Service Manual 5.3
E5	AC overcurrent	Service Manual 5.2

> Load the Service Manual to complete any diagnostic procedure.